DQN vs DQN - Stat Tests

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1 Imports

[114]:
```
import numpy as np
import pandas as pd
from scipy.stats import kruskal
import scikit_posthocs as sp
```

2 Logit, Firm 1

2.1 Data Loading

[115]:
```
model = 'logit' # row number = 0

avg_prices = {
    "shock_none": [],
    "shock_a": [],
    "shock_b": [],
    "shock_c": []
}

deltas = {
    "shock_none": [],
    "shock_a": [],
    "shock_b": [],
    "shock_c": []
}

rpdis = {
    "shock_none": [],
    "shock_a": [],
    "shock_b": [],
    "shock_c": []
}

for i in range(1, 8):
    # load the dataframes
    df_shock_none = pd.read_csv(f"../results/run_{i}/shock_none/dqn_vs_dqn.csv")
```

```
    df_shock_a = pd.read_csv(f"../results/run_{i}/shock_a/dqn_vs_dqn_schemeA.
↪csv")
    df_shock_b = pd.read_csv(f"../results/run_{i}/shock_b/dqn_vs_dqn_schemeB.
↪csv")
    df_shock_c = pd.read_csv(f"../results/run_{i}/shock_c/dqn_vs_dqn_schemeC.
↪csv")

    # average prices
    avg_prices["shock_none"].append(df_shock_none.iat[0, 1])
    avg_prices["shock_a"].append(df_shock_a.iat[0, 1])
    avg_prices["shock_b"].append(df_shock_b.iat[0, 1])
    avg_prices["shock_c"].append(df_shock_c.iat[0, 1])

    # deltas
    deltas["shock_none"].append(df_shock_none.iat[0, 6])
    deltas["shock_a"].append(df_shock_a.iat[0, 6])
    deltas["shock_b"].append(df_shock_b.iat[0, 6])
    deltas["shock_c"].append(df_shock_c.iat[0, 6])

    # rpdis
    rpdis["shock_none"].append(df_shock_none.iat[0, 8])
    rpdis["shock_a"].append(df_shock_a.iat[0, 8])
    rpdis["shock_b"].append(df_shock_b.iat[0, 8])
    rpdis["shock_c"].append(df_shock_c.iat[0, 8])
```

2.2 Kruskal-Wallis Test - Average Prices

[116]:
```
h_stat, p_val = kruskal(
    avg_prices["shock_none"],
    avg_prices["shock_a"],
    avg_prices["shock_b"],
    avg_prices["shock_c"]
)
n = 4 * 7 # total number of observations across all groups
epsilon_squared = h_stat / (n - 1)

print("Kruskal-Wallis Test on Average Prices")
print(f"H-statistic: {h_stat:.2f}, P-value: {p_val:.8f}, epsilon_squared:␣
↪{epsilon_squared:.4f}\n")
```

```
Kruskal-Wallis Test on Average Prices
H-statistic: 27.00, P-value: 0.00000589, epsilon_squared: 1.0000
```

2.3 Post-Hoc Testing - Average Prices

[117]:
```
dunn_matrix = sp.posthoc_dunn(
    [
        avg_prices["shock_none"],
        avg_prices["shock_a"],
        avg_prices["shock_b"],
        avg_prices["shock_c"]
    ], p_adjust='holm'
)

print(dunn_matrix)
```

```
          1         2         3         4
1  1.000000  0.001431  1.000000  0.001431
2  0.001431  1.000000  0.001431  1.000000
3  1.000000  0.001431  1.000000  0.001431
4  0.001431  1.000000  0.001431  1.000000
```

Every configuration produces different average prices which are significantly different from each other.

2.4 Kruskal-Wallis Test - Deltas

[118]:
```
h_stat, p_val = kruskal(
    deltas["shock_none"],
    deltas["shock_a"],
    deltas["shock_b"],
    deltas["shock_c"]
)
n = 4 * 7 # total number of observations across all groups
epsilon_squared = h_stat / (n - 1)

print("Kruskal-Wallis Test on Deltas")
print(f"H-statistic: {h_stat:.2f}, P-value: {p_val:.8f}, epsilon_squared:␣
↪{epsilon_squared:.4f}\n")
```

```
Kruskal-Wallis Test on Deltas
H-statistic: 25.61, P-value: 0.00001150, epsilon_squared: 0.9486
```

2.5 Post-Hoc Testing - Deltas

[119]:
```
dunn_matrix = sp.posthoc_dunn(
    [
        deltas["shock_none"],
        deltas["shock_a"],
        deltas["shock_b"],
        deltas["shock_c"]
```

```
    ], p_adjust='holm'
)

print(dunn_matrix)
```

```
          1         2         3         4
1  1.000000  0.006855  0.328563  0.000009
2  0.006855  1.000000  0.328563  0.328563
3  0.328563  0.328563  1.000000  0.006855
4  0.000009  0.328563  0.006855  1.000000
```

- `deltas["shock_none"]` is significantly different from `deltas["shock_a"]`
- `deltas["shock_none"]` is significantly different from `deltas["shock_c"]`
- `deltas["shock_b"]` is significantly different from `deltas["shock_c"]`

2.6 Kruskal-Wallis Test - RPDIs

[120]:
```
h_stat, p_val = kruskal(
    rpdis["shock_none"],
    rpdis["shock_a"],
    rpdis["shock_b"],
    rpdis["shock_c"]
)
n = 4 * 7 # total number of observations across all groups
epsilon_squared = h_stat / (n - 1)

print("Kruskal-Wallis Test on RPDIs")
print(f"H-statistic: {h_stat:.2f}, P-value: {p_val:.8f}, epsilon_squared: 
 ↪{epsilon_squared:.4f}\n")
```

```
Kruskal-Wallis Test on RPDIs
H-statistic: 25.61, P-value: 0.00001150, epsilon_squared: 0.9486
```

2.7 Post-Hoc Testing - RPDIs

[121]:
```
dunn_matrix = sp.posthoc_dunn(
    [
        rpdis["shock_none"],
        rpdis["shock_a"],
        rpdis["shock_b"],
        rpdis["shock_c"]
    ], p_adjust='holm'
)

print(dunn_matrix)
```

```
          1         2         3         4
1  1.000000  0.006855  0.328563  0.000009
```

```
2  0.006855  1.000000  0.328563  0.328563
3  0.328563  0.328563  1.000000  0.006855
4  0.000009  0.328563  0.006855  1.000000
```

- `rpdis["shock_none"]` is significantly different from `rpdis["shock_a"]`
- `rpdis["shock_none"]` is significantly different from `rpdis["shock_c"]`
- `rpdis["shock_b"]` is significantly different from `rpdis["shock_c"]`

3 Hotelling, Firm 1

3.1 Data Loading

[122]:
```
model = 'hotelling' # row number = 1

avg_prices = {
    "shock_none": [],
    "shock_a": [],
    "shock_b": [],
    "shock_c": []
}

deltas = {
    "shock_none": [],
    "shock_a": [],
    "shock_b": [],
    "shock_c": []
}

rpdis = {
    "shock_none": [],
    "shock_a": [],
    "shock_b": [],
    "shock_c": []
}

for i in range(1, 8):
    # load the dataframes
    df_shock_none = pd.read_csv(f"../results/run_{i}/shock_none/dqn_vs_dqn.csv")
    df_shock_a = pd.read_csv(f"../results/run_{i}/shock_a/dqn_vs_dqn_schemeA.
↪csv")
    df_shock_b = pd.read_csv(f"../results/run_{i}/shock_b/dqn_vs_dqn_schemeB.
↪csv")
    df_shock_c = pd.read_csv(f"../results/run_{i}/shock_c/dqn_vs_dqn_schemeC.
↪csv")

    # average prices
    avg_prices["shock_none"].append(df_shock_none.iat[1, 1])
```

```
    avg_prices["shock_a"].append(df_shock_a.iat[1, 1])
    avg_prices["shock_b"].append(df_shock_b.iat[1, 1])
    avg_prices["shock_c"].append(df_shock_c.iat[1, 1])

    # deltas
    deltas["shock_none"].append(df_shock_none.iat[1, 6])
    deltas["shock_a"].append(df_shock_a.iat[1, 6])
    deltas["shock_b"].append(df_shock_b.iat[1, 6])
    deltas["shock_c"].append(df_shock_c.iat[1, 6])

    # rpdis
    rpdis["shock_none"].append(df_shock_none.iat[1, 8])
    rpdis["shock_a"].append(df_shock_a.iat[1, 8])
    rpdis["shock_b"].append(df_shock_b.iat[1, 8])
    rpdis["shock_c"].append(df_shock_c.iat[1, 8])
```

3.2 Kruskal-Wallis Test - Average Prices

[123]:
```
h_stat, p_val = kruskal(
    avg_prices["shock_none"],
    avg_prices["shock_a"],
    avg_prices["shock_b"],
    avg_prices["shock_c"]
)
n = 4 * 7 # total number of observations across all groups
epsilon_squared = h_stat / (n - 1)

print("Kruskal-Wallis Test on Average Prices")
print(f"H-statistic: {h_stat:.2f}, P-value: {p_val:.8f}, epsilon_squared:␣
↪{epsilon_squared:.4f}\n")
```

```
Kruskal-Wallis Test on Average Prices
H-statistic: 15.63, P-value: 0.00134925, epsilon_squared: 0.5789
```

3.3 Post-Hoc Testing - Average Prices

[124]:
```
dunn_matrix = sp.posthoc_dunn(
    [
        avg_prices["shock_none"],
        avg_prices["shock_a"],
        avg_prices["shock_b"],
        avg_prices["shock_c"]
    ], p_adjust='holm'
)

print(dunn_matrix)
```

	1	2	3	4
1	1.000000	0.367295	1.000000	0.004482
2	0.367295	1.000000	0.367295	0.367295
3	1.000000	0.367295	1.000000	0.004482
4	0.004482	0.367295	0.004482	1.000000

- `avg_prices["shock_none"]` is significantly different from `avg_prices["shock_c"]`
- `avg_prices["shock_b"]` is significantly different from `avg_prices["shock_c"]`

3.4 Kruskal-Wallis Test - Deltas

[125]:
```python
h_stat, p_val = kruskal(
    deltas["shock_none"],
    deltas["shock_a"],
    deltas["shock_b"],
    deltas["shock_c"]
)
n = 4 * 7 # total number of observations across all groups
epsilon_squared = h_stat / (n - 1)

print("Kruskal-Wallis Test on Deltas")
print(f"H-statistic: {h_stat:.2f}, P-value: {p_val:.8f}, epsilon_squared:␣
↪{epsilon_squared:.4f}\n")
```

```
Kruskal-Wallis Test on Deltas
H-statistic: 13.71, P-value: 0.00333278, epsilon_squared: 0.5077
```

3.5 Post-Hoc Testing - Deltas

[126]:
```python
dunn_matrix = sp.posthoc_dunn(
    [
        deltas["shock_none"],
        deltas["shock_a"],
        deltas["shock_b"],
        deltas["shock_c"]
    ], p_adjust='holm'
)

print(dunn_matrix)
```

	1	2	3	4
1	1.000000	0.125565	0.717787	0.008147
2	0.125565	1.000000	0.220044	0.586153
3	0.717787	0.220044	1.000000	0.022412
4	0.008147	0.586153	0.022412	1.000000

- `deltas["shock_none"]` is significantly different from `deltas["shock_c"]`
- `deltas["shock_b"]` is significantly different from `deltas["shock_c"]`

3.6 Kruskal-Wallis Test - RPDIs

[127]:
```
h_stat, p_val = kruskal(
    rpdis["shock_none"],
    rpdis["shock_a"],
    rpdis["shock_b"],
    rpdis["shock_c"]
)
n = 4 * 7 # total number of observations across all groups
epsilon_squared = h_stat / (n - 1)

print("Kruskal-Wallis Test on RPDIs")
print(f"H-statistic: {h_stat:.2f}, P-value: {p_val:.8f}, epsilon_squared:␣
↪{epsilon_squared:.4f}\n")
```

```
Kruskal-Wallis Test on RPDIs
H-statistic: 20.94, P-value: 0.00010843, epsilon_squared: 0.7755
```

3.7 Post-Hoc Testing - RPDIs

[128]:
```
dunn_matrix = sp.posthoc_dunn(
    [
        rpdis["shock_none"],
        rpdis["shock_a"],
        rpdis["shock_b"],
        rpdis["shock_c"]
    ], p_adjust='holm'
)

print(dunn_matrix)
```

```
          1         2         3         4
1  1.000000  0.037078  0.633957  0.020554
2  0.037078  1.000000  0.002305  0.766848
3  0.633957  0.002305  1.000000  0.000872
4  0.020554  0.766848  0.000872  1.000000
```

- `rpdis["shock_none"]` is significantly different from `rpdis["shock_a"]`
- `rpdis["shock_none"]` is significantly different from `rpdis["shock_c"]`
- `rpdis["shock_a"]` is significantly different from `rpdis["shock_b"]`
- `rpdis["shock_b"]` is significantly different from `rpdis["shock_c"]`

4 Linear, Firm 1

4.1 Data Loading

[129]:
```
model = 'linear' # row number = 2

avg_prices = {
    "shock_none": [],
    "shock_a": [],
    "shock_b": [],
    "shock_c": []
}

deltas = {
    "shock_none": [],
    "shock_a": [],
    "shock_b": [],
    "shock_c": []
}

rpdis = {
    "shock_none": [],
    "shock_a": [],
    "shock_b": [],
    "shock_c": []
}

for i in range(1, 8):
    # load the dataframes
    df_shock_none = pd.read_csv(f"../results/run_{i}/shock_none/dqn_vs_dqn.csv")
    df_shock_a = pd.read_csv(f"../results/run_{i}/shock_a/dqn_vs_dqn_schemeA.
 ↪csv")
    df_shock_b = pd.read_csv(f"../results/run_{i}/shock_b/dqn_vs_dqn_schemeB.
 ↪csv")
    df_shock_c = pd.read_csv(f"../results/run_{i}/shock_c/dqn_vs_dqn_schemeC.
 ↪csv")

    # average prices
    avg_prices["shock_none"].append(df_shock_none.iat[2, 1])
    avg_prices["shock_a"].append(df_shock_a.iat[2, 1])
    avg_prices["shock_b"].append(df_shock_b.iat[2, 1])
    avg_prices["shock_c"].append(df_shock_c.iat[2, 1])

    # deltas
    deltas["shock_none"].append(df_shock_none.iat[2, 6])
    deltas["shock_a"].append(df_shock_a.iat[2, 6])
    deltas["shock_b"].append(df_shock_b.iat[2, 6])
```

```
    deltas["shock_c"].append(df_shock_c.iat[2, 6])

    # rpdis
    rpdis["shock_none"].append(df_shock_none.iat[2, 8])
    rpdis["shock_a"].append(df_shock_a.iat[2, 8])
    rpdis["shock_b"].append(df_shock_b.iat[2, 8])
    rpdis["shock_c"].append(df_shock_c.iat[2, 8])
```

4.2 Kruskal-Wallis Test - Average Prices

[130]:
```
try:
    h_stat, p_val = kruskal(
        avg_prices["shock_none"],
        avg_prices["shock_a"],
        avg_prices["shock_b"],
        avg_prices["shock_c"]
    )

    n = 4 * 7 # total number of observations across all groups
    epsilon_squared = h_stat / (n - 1)

    print("Kruskal-Wallis Test on Average Prices")
    print(f"H-statistic: {h_stat:.2f}, P-value: {p_val:.8f}, epsilon_squared:␣
 ↪{epsilon_squared:.4f}\n")
except ValueError:
    print("All numbers are identical in kruskal")

    print(f"avg_prices[\"shock_none\"] = {[float(x) for x in␣
 ↪avg_prices["shock_none"]]}")
    print(f"avg_prices[\"shock_a\"] = {[float(x) for x in␣
 ↪avg_prices["shock_a"]]}")
    print(f"avg_prices[\"shock_b\"] = {[float(x) for x in␣
 ↪avg_prices["shock_b"]]}")
    print(f"avg_prices[\"shock_c\"] = {[float(x) for x in␣
 ↪avg_prices["shock_c"]]}")
```

```
All numbers are identical in kruskal
avg_prices["shock_none"] = [0.47, 0.47, 0.47, 0.47, 0.47, 0.47, 0.47]
avg_prices["shock_a"] = [0.47, 0.47, 0.47, 0.47, 0.47, 0.47, 0.47]
avg_prices["shock_b"] = [0.47, 0.47, 0.47, 0.47, 0.47, 0.47, 0.47]
avg_prices["shock_c"] = [0.47, 0.47, 0.47, 0.47, 0.47, 0.47, 0.47]
```

There are are no differences between any of the average prices.

4.3 Post-Hoc Testing - Average Prices

Could not be performed.

4.4 Kruskal-Wallis Test - Deltas

[131]:
```
h_stat, p_val = kruskal(
    deltas["shock_none"],
    deltas["shock_a"],
    deltas["shock_b"],
    deltas["shock_c"]
)
n = 4 * 7 # total number of observations across all groups
epsilon_squared = h_stat / (n - 1)

print("Kruskal-Wallis Test on Deltas")
print(f"H-statistic: {h_stat:.2f}, P-value: {p_val:.8f}, epsilon_squared:␣
 ↪{epsilon_squared:.4f}\n")
```

```
Kruskal-Wallis Test on Deltas
H-statistic: 24.08, P-value: 0.00002404, epsilon_squared: 0.8918
```

4.5 Post-Hoc Testing - Deltas

[132]:
```
dunn_matrix = sp.posthoc_dunn(
    [
        deltas["shock_none"],
        deltas["shock_a"],
        deltas["shock_b"],
        deltas["shock_c"]
    ], p_adjust='holm'
)

print(dunn_matrix)
```

```
          1         2         3         4
1  1.000000  0.022021  0.487700  0.000066
2  0.022021  1.000000  0.112051  0.210779
3  0.487700  0.112051  1.000000  0.001073
4  0.000066  0.210779  0.001073  1.000000
```

- `deltas["shock_none"]` is significantly different from `deltas["shock_a"]`
- `deltas["shock_none"]` is significantly different from `deltas["shock_c"]`
- `deltas["shock_b"]` is significantly different from `deltas["shock_c"]`

4.6 Kruskal-Wallis Test - RPDIs

[133]:
```
h_stat, p_val = kruskal(
    rpdis["shock_none"],
    rpdis["shock_a"],
    rpdis["shock_b"],
    rpdis["shock_c"]
```

```
)
n = 4 * 7 # total number of observations across all groups
epsilon_squared = h_stat / (n - 1)

print("Kruskal-Wallis Test on RPDIs")
print(f"H-statistic: {h_stat:.2f}, P-value: {p_val:.8f}, epsilon_squared:␣
↪{epsilon_squared:.4f}\n")
```

```
Kruskal-Wallis Test on RPDIs
H-statistic: 19.67, P-value: 0.00019854, epsilon_squared: 0.7286
```

4.7 Post-Hoc Testing - RPDIs

[134]:
```
dunn_matrix = sp.posthoc_dunn(
    [
        rpdis["shock_none"],
        rpdis["shock_a"],
        rpdis["shock_b"],
        rpdis["shock_c"]
    ], p_adjust='holm'
)

print(dunn_matrix)
```

```
          1         2         3         4
1  1.000000  0.003815  1.000000  0.003815
2  0.003815  1.000000  0.020775  1.000000
3  1.000000  0.020775  1.000000  0.020775
4  0.003815  1.000000  0.020775  1.000000
```

- `rpdis["shock_none"]` is significantly different from `rpdis["shock_a"]`
- `rpdis["shock_none"]` is significantly different from `rpdis["shock_c"]`
- `rpdis["shock_a"]` is significantly different from `rpdis["shock_b"]`
- `rpdis["shock_b"]` is significantly different from `rpdis["shock_c"]`